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Bill of Materials

1. 4-pin GNSS – GG-1802 GPS module (NEO-6M) – \$4 to \$5 a piece
2. 8-pin, 3.65cm TFT ILI9163C display – \$3 a piece
3. ESP-32S board – \$4 to \$7 [Depending upon which model you buy, the bare minimum is \$4 while the USB programmable is \$7]
4. Toggle on/off switch – 1 piece
5. 3.7V battery – 1 piece
6. 470μF, 16V electrolytic capacitor – 1 piece
7. Some jumper wires for connections

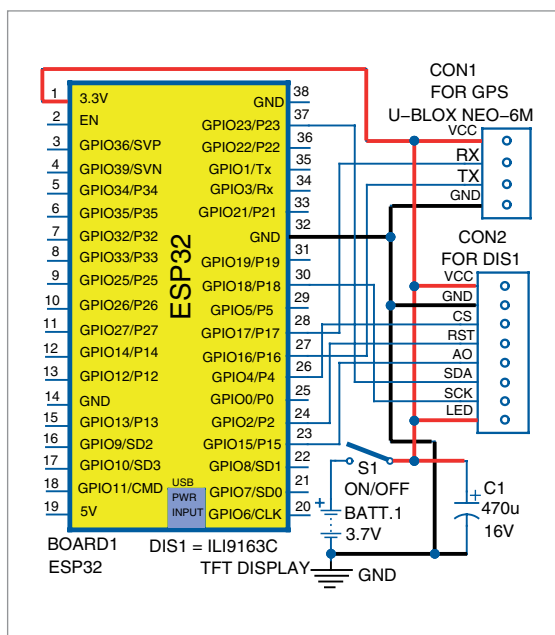


Fig. 5: Circuit diagram of ESP32 based GPS receiver

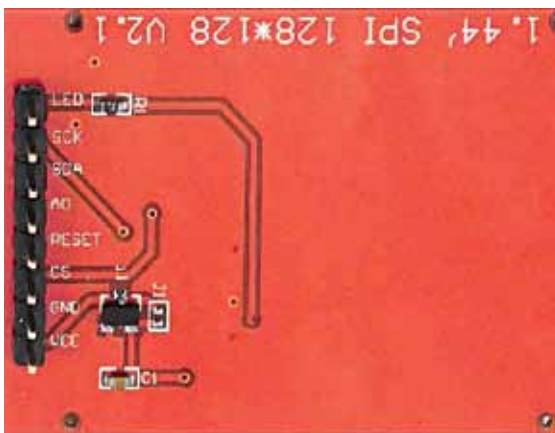


Fig. 6: Pin details of TFT display

You need header files such as *Adafruit_GFX_Library* and *TFT_ILI9163C* to successfully compile and run the code of this project.

Aftermath

With restoration of these cheap GPS modules by reading their UART data, it was rather easy to decipher them rather than using the fancy ready-made programs that the Arduino galaxy is replete with such as GPS, Adafruit-GPS, NEOGPS, TinyGPS, TinyGPS ++, and NEMAGPSc. So, I can now go ahead with the gas pipeline detection project I have in mind with the spare GPS module and, whenever required, can peep into more UART devices in future—before looking out for readymade header programs.

Finally, I called the guy at *aliexpress.com* so that I could pay back the amount they had refunded me on my lodging the complaint. All they sent me was a big thank-you note and requested me to send them the Arduino sketch! **EFY**

EFY Note
The source code
of this project
is available
for free download
at source.efymag.com



Somnath Bera is an avid user of open source software. Professionally, he is a thermal power expert and works as additional general manager at NTPC Ltd